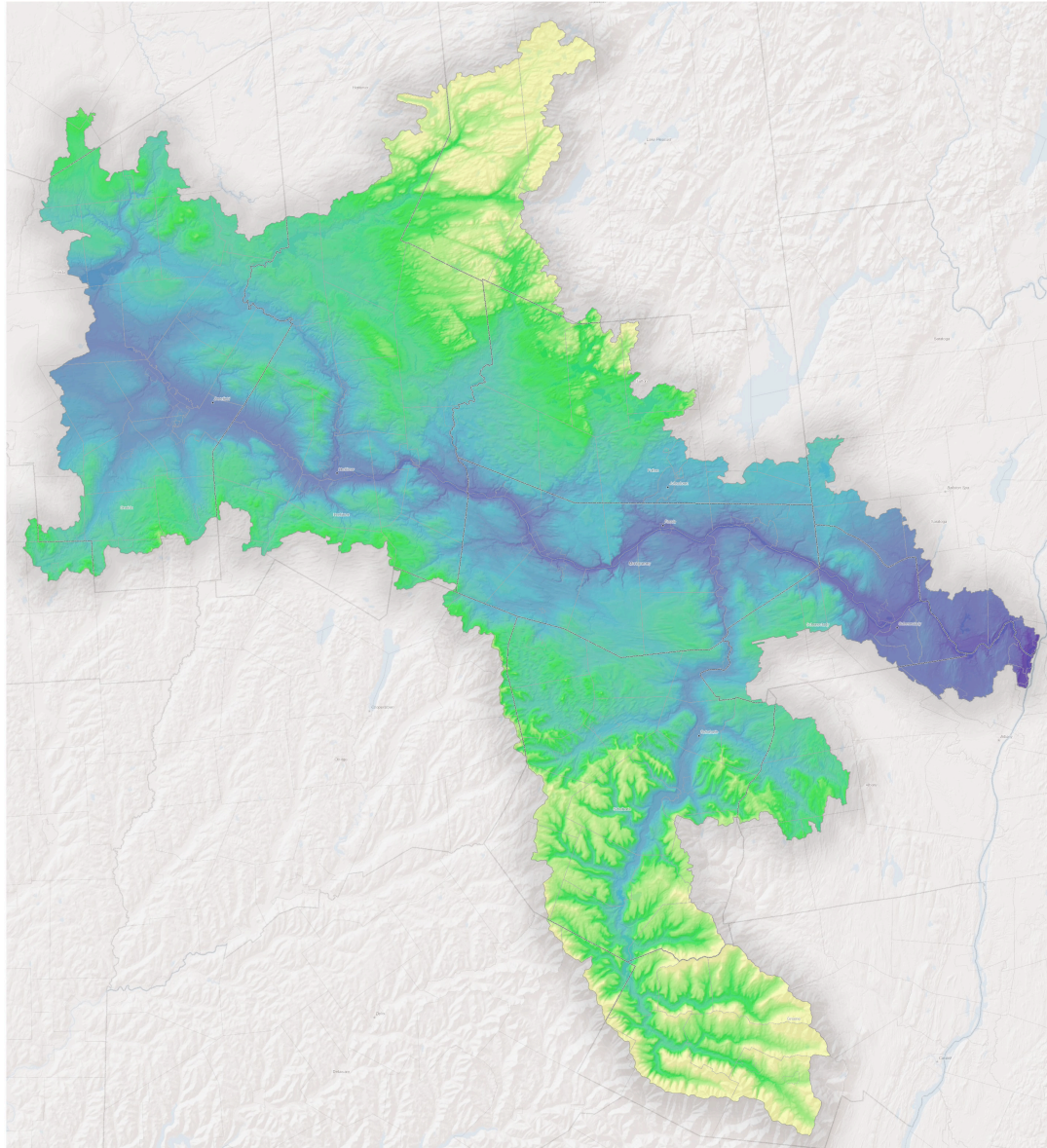


Mohawk Watershed Symposium 2014



Abstracts and program

College Park Hall, Union College
Schenectady NY
21 March 2014

PREDICTION MODEL FOR THE WATER DISCHARGE TIME SERIES IN MOHAWK WATERSHED, NY

Antonios Marsellos¹ and Katerina Tsakiri²

¹School of Environment and Technology, University of Brighton, BN2 4GJ, United Kingdom

²School of Computing, Engineering, and Mathematics, University of Brighton, BN2 4GJ, United Kingdom

Introduction

There is a connection between an increase in flooding and the climatic change (Bronstert 1995; Kotlarski et al. 2012). In particular, storms with increasing amount of moisture produce more intense precipitation events therefore increasing the risk of flooding (Trenberth 2011). Temperature increases during the winter period can break-up the river's icy surface and hence initiate "ice jams". In the last decade, Mohawk River has shown a significant increase in water discharge time series (Kern, A., 2008; Marsellos et al. 2010a; Marsellos et al. 2010b; Tsakiri et al. 2014). Those events may be associated with the different climatic conditions.

In this study, we describe a model for the explanation and prediction of the daily water discharge time series derived by three locations nearby the Mohawk River, NY (Fig. 1) during the period 2005-2013. For the analysis of the model, we use the daily water discharge time series, the daily data of ground water level and the climatic variables from a nearby location.

Methodology

For our study, we use a statistical methodology to decompose the time series of all the variables into different components (long, seasonal and short term component). The long-term component describes the fluctuations of a time series defined as being longer than a given threshold; the seasonal component describes the year-to-year fluctuations, while the short term component describes the short term variations. The Kolmogorov-Zurbenko (KZ) filter is used for the decomposition of the time series (Zurbenko 1986). The KZ filter, which separates the long term variations from the short term variations in a time series (Yang and Zurbenko 2010a) provides a simple design and the smallest level of interferences between scales (long, seasonal and short term components) of a time series. Moreover the KZ filter can be applied directly to datasets containing missing observations (Eskridge et al. 1997; Rao et al. 1997; Yang and Zurbenko, 2010b).

In our analysis, we first decompose the time series of the water discharge derived by three different locations nearby the Mohawk River and then we predict each component of the water discharge time series separately by using the ground water level and the climatic variables. By using the decomposition of the time series, the explanation of the water discharge time series has been improved approximately two times compared to the raw data. Figure 1 shows three examples of the prediction of the water discharge time series for three different water discharge monitoring stations in the Mohawk watershed.

Conclusions

The decomposition of the time series reveals different correlation patterns between the components of the time series. Moreover, the isolation of the short-term variations from the time series shows a summer period with water table replenishment and a winter period with water table depletion. The design of multivariate models as well as the decomposition of the time series improves the prediction of flooding caused by storms, rapid snowmelt and ice jams.

References

- Bronstert, A. (1995) River Flooding in Germany: Influence by Climate Change? *Physics and Chemistry of the Earth*, 20, No 6, 445-450.
- Eskridge, R.E., Ku J.U., Porter, P.S., Rao S.T., Zurbenko, I., (1997) Separating Different Scales of Motion in Time Series of Meteorological Variables. *Bull Am Meteorol Soc* 78:1473-1483.
- Kern, A.L., 2008. Study of 20th century trends in stream flow for West Canada and Schoharie Creeks of the Mohawk-Hudson Rivers watershed, Department of Geology, Union College, Schenectady, New York, thesis.
- Kotlarski, S., Hagamann, S., Krahe, P., Podzun, R., Jacob, D., (2012), The Elbe river flooding 2002 as seen by an extended regional climate model, *Journal of Hydrology*, Vol 472-473, 169-183
- Marsellos, A.E., Garver, J.I., Cockburn, J.M.H., Tsakiri, K., (2010a) Determination of historical channel changes and meander cut-off points using

- LiDAR and GIS in Schoharie creek, NY, Proceedings from the Mohawk Symposium 2010, in Schenectady, New York, 28-33.
- Marsellos, A.E. and Garver, J.I., (2010b) Channel incision and landslides identified by LiDAR in the lower reaches of Schoharie creek, New York, Geol. Soc. Am., 42, 1, 73.
- Rao, S.T., Zurbenko, I.G., Neagu, R., Porter, P.S., Ku J.Y., and Henry, R.F. (1997) Space and time scales in ambient ozone data. Bull Am Meteorol Soc 78:2153-2166.
- Tsakiri, K.T., Marsellos, A.M., Zurbenko, I.G. (2014). An efficient prediction model for water discharge in Schoharie Creek, NY. Journal of Climatology, vol 2014, Article ID 284137, 10 pages, 2014. doi:10.1155/2014/284137.
- Yang, W., and Zurbenko, I. (2010a) Kolmogorov-Zurbenko filters, Wiley Interdisciplinary Reviews. In: Computational Statistics, Wiley 2:340-351. doi: <http://wires.wiley.com/WileyCDA/WiresArticle/wisId-WICS71.html>.
- Yang, W., and Zurbenko, I. (2010b) Nonstationarity Wiley Interdisciplinary Reviews. In: Computational Statistics, Wiley 2:107-115.
- Zurbenko, I.G., (1986) The Spectral Analysis of Time Series. Amsterdam, North Holland Series in Statistics and Probability.

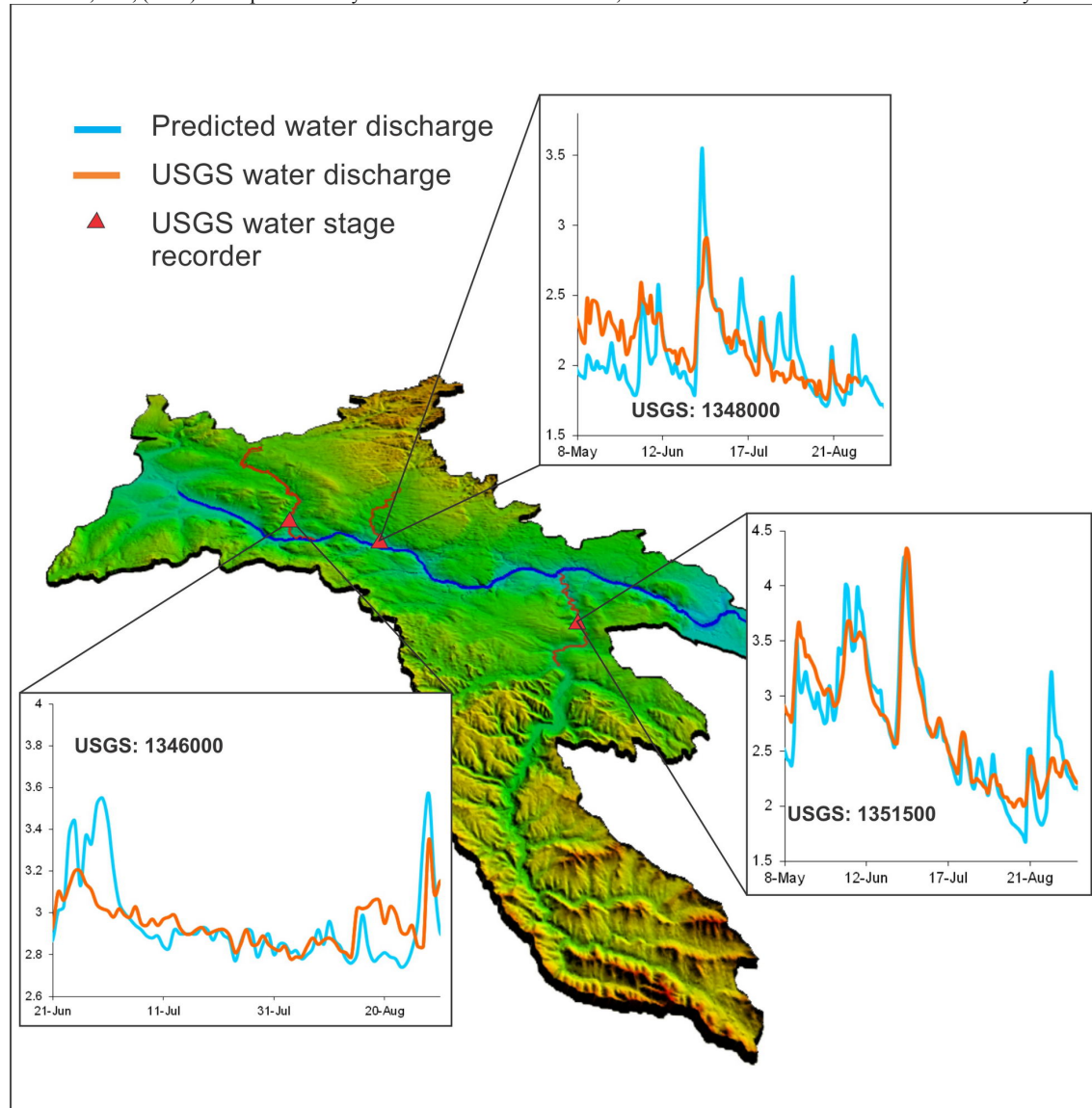


Figure 1: Raw water discharge data (blue line) along with the prediction model (orange line) for three locations nearby the Mohawk River.

Poster Presentation